

local-1777891131920

Comprehensive Hardware Design Analysis

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Executive Summary

Generated

Project ID	local-1777891131920
Industry Domain	hvac_commercial_heating
Engine Version	PDF Engine v2.1 (Full Detail)
Target Quantity	Not specified

ARCHITECTURE AT A GLANCE

MODULES	BOM ROWS	IDENTIFIED RISKS
10	48	30

ECONOMICS

EST. UNIT COST	TARGET CEILING	HEADROOM
£9,367.50	None	N/A

TOTAL NRE

£6,000.00

UNIT + NRE AMORTISED

£15,367.50

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1. Brief & Requirements

1.1 Overview & Context

Subject / Use Case

Providing space heating and low-temperature hot water to commercial properties under 1,000 sqm as a standalone drop-in replacement for medium-sized aging gas boilers.

Mission Statement

To accelerate the decarbonization of small commercial real estate by providing highly efficient, low-GWP, plug-and-play R290 heat pumps that simplify systemic electrification.

Target Customers

Small commercial building owners, HVAC cascade retro-fit contractors, retail franchise operators, medium-level facility managers, boutiques, and clinics.

Why Now (Market Timing)

Stringent global F-Gas phasedowns have accelerated the need for pure natural refrigerants, while commercial soaring energy costs and ESG mandates push operators away from fossil fuel boilers.

Full Synthesis Report

The development of a 30kW commercial air-source heat pump (ASHP) utilizing R290 (propane) refrigerant is highly timely, addressing a critical intersection of deep decarbonization goals and stringent global regulatory shifts. The revised EU F-Gas Regulation and the US EPA AIM Act are rapidly phasing out legacy high-Global Warming Potential (GWP) refrigerants. With a negligible GWP of 3, R290 provides absolute future-proofing against these bans. From a thermodynamic perspective, R290 is superior to typical HFCs, easily supporting the target Coefficient of Performance (COP) of e 3.5 at A2/W35 and allowing for flow temperatures up to 75°C, making it exceptionally versatile for retrofitting older, radiator-heated commercial spaces without needing full envelope upgrades.

However, reaching a 30kW capacity exclusively in a single unit presents considerable engineering complexities. R290 is an A3 (highly flammable) safety class refrigerant. A 30kW system requires a larger refrigerant charge footprint, engaging stricter safety protocols under standards like EN 378 and IEC 60335-2-40. The design must be a strict monobloc configuration, keeping all volatile refrigerants outside the building envelope and circulating only water or a glycol mix indoors. This necessitates hermetically sealed components, ATEX-rated explosion-proof contactors, and meticulous active ventilation on the external unit.

The specific market segment of small commercial buildings (e.g., boutique retail, medium-sized offices, small clinics) is chron-

ically underserved by right-sized, plug-and-play systems. Many installers resort to cascading multiple smaller residential heat pumps to reach 30kW, which increases spatial footprint, hydraulic complexity, and installation costs. Producing a native, compact 30kW R290 unit will provide a highly sought-after, efficient drop-in replacement for retiring commercial gas boilers.

1.2 Market & Competitors

Competitor Landscape

Viessmann

Product	Vitocal 250-A Pro
Pricing Strategy	\$18,000 - \$25,000 depending on control module and options
Technical Specs	R290 refrigerant, up to 39.5kW heating capacity, large monobloc format, max flow temperatures up to 70°C.

STRENGTHS

Premium brand trust, highly advanced digital diagnostic controls, excellent build quality and low noise.

WEAKNESSES

High capital cost, industrial-grade dimensions requiring significant footprint real estate.

DIFFERENTIATION ANGLE

Engineer a more compact structural frame allowing placement in tighter urban commercial alleyways or smaller side plots.

Clivet

Product	Thunder R290 (Air-to-Water MT)
Pricing Strategy	\$16,000 - \$22,000
Technical Specs	R290, modular output covering the 40kW scale, Eurovent certified, variable speed scroll compressor.

STRENGTHS

Modular cascading allows up to 16 joined units, robust commercial-first structural integrity, good BMS integration.

WEAKNESSES

Heavy, utilitarian aesthetics lacking subtle architectural value for visible retail settings; often overpowered for simpler small needs.

DIFFERENTIATION ANGLE

Outperform on part-load A2/W35 efficiency and prioritize highly aesthetic, architect-friendly casing designs for street-visible setups.

Vaillant

Product	aroTHERM plus (Cascaded configuration)
Pricing Strategy	\$22,000+ (for three individual units plus cascade hydraulic controllers)
Technical Specs	R290 refrigerant, typically utilizing a 3 x 12kW cascade array to approximate ~36kW total, COP ~3.6 at A2/W35.

STRENGTHS

Highly familiar to installers, unparalleled system reliability, excellent intelligent control interface.

WEAKNESSES

Requires three independent units, tripling the points of mechanical failure, taking up massive wall space, and increasing piping complexity.

DIFFERENTIATION ANGLE

Provide a monolithic all-in-one 30kW unit that significantly slashes installation man-hours and hydraulic complexity compared to a tri-unit cascade.

1.3 Engineering Constraints

Core Targets

Unit Cost Ceiling	£0.00
Max Mass	-
Target Process	
Target Material	
Tolerance Target	
Production Quantity	

Research Sources

TITLE	TYPE	YEAR	RELEVANCE
European F-Gas Regulation (EU) 2024/573	Regulation	2026	Dictates the timeline for the phase-out of high-GWP refrigerants, validating the necessity of an R290 solution.
EN 378-1:2016 / ISO 5149	Safety Standard	2026	Outlines safety and environmental strictness, directly driving the allowable refrigerant charge and zone mitigations for R290.
EHPA European Heat Pump Market and Statistics Re-Market Report		2026	Identifies the growth delta and capacity demand in the light commercial sector, supporting the 30kW usecase.
ASHRAE Guideline 15	Safety Guideline	2026	Crucial for mapping out the mechanical room and outdoor limits for highly flammable A3 refrigerants if selling to US markets.
EcoDesign Directive (ErP) Lot 1	Efficiency Policy	2026	Mandates the minimum seasonal efficiency (SCOP) and acoustic output bounds required for legal CE marking and sales.

2.1 EN 378-1:2016

Refrigerating systems and heat pumps - Safety and environmental requirements

Status: not-started

Owner: Lead Mechanical Engineer

Standard Summary

Specifies charge limits and safety requirements for systems using A3 (highly flammable) refrigerants.

Applicability Assessment

EU/International

Engineering Design Impact

Requires precise calculation of max R290 charge and definition of ATEX zones/exclusion radiuses around the monobloc unit.

EVIDENCE REQUIRED

Safety dossier, charge calculation report, standardized risk assessment.

GAP ACTION

Complete theoretical R290 charge size limits based on preliminary piping geometry.

2.2 IEC 60335-2-40

Particular requirements for electrical heat pumps, air-conditioners and dehumidifiers

Status: not-started

Owner: Lead Electrical Engineer

Standard Summary

Specific electrical and functional safety limits for systems containing flammable refrigerants.

Applicability Assessment

Global

Engineering Design Impact

All electrical components near the compressor (contactors, relays) must be certified non-sparking, or hermetically separated from potential leak zones.

EVIDENCE REQUIRED

CB test report from independent lab, leak simulation test approvals.

GAP ACTION

Audit supplier list for ATEX/non-sparking rated components for the main electrical chassis.

2.3 ErP Directive 2009/125/EC

Ecodesign requirements for space heaters

Status: not-started

Owner: Systems & Thermal Engineer

Standard Summary

Mandates European minimum seasonal energy efficiency (SCOP) metrics and sound power level limits.

Applicability Assessment

EU

Engineering Design Impact

Fan blade speed, compressor mapping, and chassis dampening must balance thermal performance (COP >= 3.5) with strict dBA noise levels.

EVIDENCE REQUIRED

Approved energy product fiche, accredited third-party acoustical and performance lab tests.

GAP ACTION

Construct a digital twin simulation for seasonal SCOP metrics and baseline acoustic output.

2.4 PED 2014/68/EU

Pressure Equipment Directive

Status: not-started

Owner: Mechanical/Piping Engineer

Standard Summary

Conformity assessment rules for stationary pressure equipment operating above 0.5 bar.

Applicability Assessment

EU

Engineering Design Impact

Sizing of the condenser, compressor, and specific brazing techniques must withstand the distinct operating pressures of R290.

EVIDENCE REQUIRED

Piping diagrams, burst test reports, materials traceability certs, CE-PED declaration.

GAP ACTION

Collect valid PED certifications systematically from the planned heat exchanger and piping vendors.

2.5 EU 2024/573

Revised F-Gas Regulation

Status: not-started

Owner: Compliance & Documentation Manager

Standard Summary

Bans use of high-GWP fluorinated greenhouse gases; impacts labeling and end-of-life recovery parameters.

Applicability Assessment

EU

Engineering Design Impact

Although R290 complies with the bans, the product must feature highly specific warning labels, venting instructions, and dedicated service points.

EVIDENCE REQUIRED

Physical labeling compliance, detailed end-of-life and maintenance manuals.

GAP ACTION

Draft external decal specifications and outline the serviceability manual.

3. Sizing & Spatial Optimisation

INFEASIBLE DESIGN ALLOCATION

The current sizing constraints are not satisfied. Module requirements exceed the available envelope.

System Envelope

KIND	RULES DOMAIN	FEASIBLE
generic_room	hvac_commercial_heating	No

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Total Internal Floor Area	25 m²
Total Internal Volume	75 m³
Available Floor Budget	21.25 m²

Module Dimension Allocations

MODULE	W × D × H (MM)	AREA M²	MOUNT	SCALE
compressor_assembly	2236 × 2236 × 2400	5	floor	mass_proportional
air_side_evaporator	2236 × 2236 × 2400	5	floor	mass_proportional
water_side_condenser	2236 × 2236 × 2400	5	floor	mass_proportional
refrigerant_management	2236 × 2236 × 2400	5	floor	mass_proportional
airflow_management	2236 × 2236 × 2400	5	floor	mass_proportional
structural_chassis	2236 × 2236 × 2400	5	floor	mass_proportional
electrical_power	2236 × 2236 × 2400	5	floor	mass_proportional

control_and_safety	2236 × 2236 × 2400	5	floor	mass_proportional
outer_enclosure	2236 × 2236 × 2400	5	floor	mass_proportional
hydraulic_pumping	2236 × 2236 × 2400	5	floor	mass_proportional

Identified Conflicts

- Total module footprint (50.0m2) exceeds envelope budget (21.3m2)

Optimisation Recommendations

- Consider a larger envelope
- Optimize module footprint

4.1 Refrigerant Compression Module

Lead Time: 14 weeks

Status: draft

Module Purpose

Compresses low-pressure R290 vapor into high-temperature, high-pressure gas to drive the thermodynamic heating cycle.

Why it matters to the system

It is the primary prime mover of the system, directly dictating both the peak thermal capacity (30kW) and the seasonal energy efficiency (SCOP) of the entire unit.

Detailed Technical Description

The heart of the heat pump, this module utilizes an advanced inverter-driven scroll compressor specifically mapped for R290 thermodynamics. It works by drawing in low-pressure, low-temperature propane vapor from the evaporator and mechanically compressing it, drastically raising its pressure and temperature. The scroll involutes must endure fluctuating mechanical loads while accurately maintaining compression ratios that allow flow temperatures up to 75°C without electric backup.

Due to the highly flammable A3 classification of R290, this module must strictly adhere to IEC 60335-2-40 safety protocols. The electrical terminals are hermetically sealed or constructed as non-sparking mechanisms to prevent ignition if an internal leak occurs. Heavy acoustic jacketing and advanced inverter modulation ensure that the compressor meets ErP Lot 1 noise directives while delivering dynamic part-load efficiencies required for a COP of ≥ 3.5 at A2/W35.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Low-pressure R290 vapor 3-phase 400V electrical power - Inverter control signals	- High-pressure, high-temperature R290 gas - Vibration - Waste heat

4.1.2 Specifications & Risk Profile

Expected Key Parts

- Inverter-driven scroll compressor
- Vibration isolation mounts
- Acoustic compressor jacket
- Discharge temperature sensor
- Sump heater

Identified Failure Modes

- Liquid slugging destroying scroll involute geometry during cold starts
- Loss of oil lubrication causing bearing seizure
- Electrical winding insulation breakdown leading to terminal short

Open Questions

- Exact acoustic resonance frequencies at part-load speeds
- Optimal sump heater wattage curve for extreme cold starts below -15°C

4.1.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
compressor_assembly-001	Inverter-driven scroll compressor	COTS	-	£800.00
compressor_assembly-002	Vibration isolation mounts	COTS	-	£40.00
compressor_assembly-003	Acoustic compressor jacket	COTS	-	£800.00
compressor_assembly-004	Discharge temperature sensor	COTS	-	£60.00
compressor_assembly-005	Sump heater	COTS	-	£40.00
Module Total Cost:				£1,740.00

4.1.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The R290 compression module design is thermodynamically viable but requires critical safety and engineering additions, specifically regarding liquid slugging prevention and A3 flammability mitigation.

Identified Issues

ENGINEERING

Missing suction accumulator in the key parts list. Liquid slugging during cold starts is a stated failure mode, and a sump heater alone is insufficient to prevent it.

Suggestion: Add a properly sized suction accumulator to the module BOM to capture liquid refrigerant before it enters the scroll compressor.

RISK

Acoustic jacketing poses a gas-trapping risk. R290 is heavier than air; if the compressor leaks, the jacket could create a localized explosive pocket around electrical components.

Suggestion: Design the acoustic jacket with bottom ventilation or use non-porous materials that prevent gas pooling, ensuring compliance with IEC 60335-2-40.

MATERIALS

High discharge temperatures required to achieve 75°C output without backup heating will severely stress standard compressor lubricants and seals.

Suggestion: Specify high-thermal-stability lubricants (e.g., specialized POE or PAG oils) and high-temperature PTFE or PEEK seals in the scroll involutes.

MANUFACTURABILITY

Stringent leak tolerances are required for A3 refrigerants, necessitating expensive helium accumulation leak testing on the production line.

Suggestion: Integrate a vacuum chamber helium leak test station into the assembly line to guarantee hermeticity prior to R290 charging.

Strategic Recommendations

- Incorporate a suction accumulator into the suction line design to mitigate the risk of liquid slugging.
- Redesign the acoustic jacket to include bottom-draining ventilation to prevent R290 gas accumulation.
- Ensure the sump heater is an intrinsically safe PTC type and the electrical terminal box is fully potted or ATEX-rated.
- Implement helium mass spectrometry leak testing in the manufacturing process with a minimum leak rate threshold of 1×10^{-5} mbar L/s.
- Specify strict durometer tolerances for the vibration isolation mounts to effectively decouple variable-frequency vibrations across the entire inverter curve.

4.2 Air-Side Heat Exchanger Module

Lead Time: 12 weeks

Status: draft

Module Purpose

Extracts low-grade sensible and latent heat from the outdoor ambient air into the R290 refrigerant.

Why it matters to the system

Determines the heat pump's ability to maintain capacity in freezing winter conditions and limits the energy wasted on defrost cycles.

Detailed Technical Description

The air-side evaporator captures low-grade thermal energy from ambient air, utilizing a robust copper-tube/aluminum-fin architecture. The fin pitch must be meticulously designed to balance maximum heat transfer against the inevitable frost accumulation that occurs in high-humidity, sub-zero European operating conditions. Optimized internal grooving in the copper tubes enhances nucleate boiling of the R290, capitalizing on its superior transport properties.

Hydrophilic coatings applied to the aluminum fins reduce the surface tension of water, facilitating rapid condensate drainage during defrost cycles. Uniform refrigerant distribution relies on carefully tuned headers to prevent dry-out zones or uneven icing. Because reversing the cycle is required for defrosting, managing the two-phase dynamics and draining water quickly prevents ice blockages and reduces downtime.

SYSTEM INPUTS	SYSTEM OUTPUTS
- High-volume ambient air - Low-pressure two-phase R290 liquid/gas mixture	- Chilled ambient air - Low-pressure R290 vapor - Defrost condensate water

4.2.2 Specifications & Risk Profile

Expected Key Parts

- Copper-tube/Aluminum-fin coil bank
- Hydrophilic fin coating
- Refrigerant distribution headers
- Defrost temperature thermistors

Identified Failure Modes

- Ice bridging between fins causing complete air-flow blockage
- Galvanic corrosion between copper tubes and aluminum fins
- Micro-leaks at header brazing joints due to thermal cycling fatigue

Open Questions

- Actual frost accumulation rate under high-humidity UK ambient conditions
- Long-term durability of hydrophilic coating against atmospheric pollutants

4.2.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
air_side_evapora- tor-001	Copper-tube/Aluminum-fin coil bank	COTS	-	£40.00
air_side_evapora- tor-002	Hydrophilic fin coating	COTS	-	£40.00
air_side_evapora- tor-003	Refrigerant distribution headers	Custom	-	£80.00
air_side_evapora- tor-004	Defrost temperature thermistors	COTS	-	£40.00
Module Total Cost:				£200.00

4.2.4 Engineering Review

Review by Fang

WARN

Verdict Summary

While the copper-tube/aluminum-fin architecture is thermodynamically optimal, the high flammability of R290 paired with the risk of thermal cycling micro-leaks necessitates stringent manufacturing and inspection controls.

Identified Issues

RISK

Micro-leaks at header brazing joints present a severe safety and ignition hazard due to the A3 flammability classification of R290.

Suggestion: Require automated induction brazing for headers to ensure consistent joint penetration, and implement 100% helium mass spectrometry leak testing on the production line.

MATERIALS

Galvanic corrosion between the copper tubes and aluminum fins can degrade performance and structural integrity over time, especially in humid or coastal environments.

Suggestion: Utilize pre-coated aluminum fin stock with an epoxy or specialized anti-corrosion primer beneath the hydrophilic layer. Maintain sharp die tolerances in the fin press to minimize bare aluminum exposure at the collar shears.

ENGINEERING

Ice bridging between fins is highly dependent on fin pitch and frosting dynamics in sub-zero Europe.

Suggestion: Specify a wider fin pitch (minimum 2.5 mm, up to 3.0 mm) or consider a variable fin pitch design (wider at the air inlet) to extend the time between defrost cycles.

MANUFACTURABILITY

Internally grooved copper tubes increase the difficulty of bending and expanding without inducing micro-fractures in the wall.

Suggestion: Control the tube expansion bullet diameter accurately to balance mechanical bond strength (thermal contact) against excessive wall thinning.

Strategic Recommendations

- Define strict QA requirements for braze joints, mandating 100% continuous helium tracer gas leak testing.
- Specify a fin pitch of at least 2.5 mm for European high-humidity, sub-zero conditions to forestall ice bridging.
- Adopt automated induction brazing or controlled-atmosphere brazing for header assemblies to mitigate thermal fatigue failure.
- Validate the structural integrity of internally grooved pre-bent tubes with burst pressure testing to account for wall thinning.
- Require a 1000-hour salt spray test qualification for the hydrophilic and anti-corrosion coated fin stock to guarantee galvanic isolation.

4.3 Water-Side Heat Exchanger Module

Lead Time: 10 weeks

Status: draft

Module Purpose

Transfers heat from the superheated R290 gas to the commercial building's hydronic heating circuit.

Why it matters to the system

Directly impacts the maximum achievable water temperature and heavily dictates the total allowed system refrigerant charge.

Detailed Technical Description

This module rejects the heat of compression and phase change into the hydronic loop utilizing a high-efficiency Brazed Plate Heat Exchanger (BPHE). Constructed from 316-grade stainless steel plates brazed with copper, the BPHE maximizes turbulent heat transfer in an incredibly compact volume. By leveraging the thermodynamic characteristics of R290, this component can reliably output flow temperatures of up to 75°C, providing the required drop-in capability for legacy commercial radiator systems.

Safety and regulatory compliance mandate strict adherence to the Pressure Equipment Directive (PED 2014/68/EU), given the high internal condensing pressures of propane. Deep subcooling is achieved through asymmetric plate geometries, which minimizes the R290 internal volume (thereby reducing the total system charge to comply with EN 378). A heavy elastomeric insulating jacket surrounds the unit to mitigate parasitic heat losses to ambient cold air.

SYSTEM INPUTS	SYSTEM OUTPUTS
- High-pressure superheated R290 gas - Return water from commercial heating loop	- High-pressure subcooled R290 liquid - Heated supply water (up to 75°C)

4.3.2 Specifications & Risk Profile

Expected Key Parts

- Stainless Steel Brazed Plate Heat Exchanger (BPHE)
- Elastomeric thermal insulation jacket
- Water-side temperature sensors
- Freeze protection trace heating

Identified Failure Modes

- Water-side freezing during unpowered sub-zero ambient conditions leading to plate rupture
- Fouling or scaling of water channels deteriorating heat transfer
- Thermal shock causing fatigue failure in brazed joints

Open Questions

- Water quality variations across target commercial installation sites
- Exact internal volume reduction possible without incurring severe pressure drops

4.3.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
water_side_condenser-001	Stainless Steel Brazed Plate Heat Exchanger (BPHE)	COTS	-	£600.00
water_side_condenser-002	Elastomeric thermal insulation jacket	COTS	-	£15.00
water_side_condenser-003	Water-side temperature sensors	COTS	-	£60.00
water_side_condenser-004	Freeze protection trace heating	COTS	-	£40.00
Module Total Cost:				£715.00

4.3.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The BPHE module utilizes industry-standard 316SS, but faces critical risks regarding unpowered freeze protection and potential copper braze corrosion from untreated legacy hydronic water.

Identified Issues

RISK

Trace heating cannot prevent water-side freezing during power outages, leading to catastrophic plate rupture and R290 leakage into the water loop.

Suggestion: Implement an unpowered mechanical anti-freeze drain valve or mandate a minimum glycol concentration in the hydronic loop.

MATERIALS

Copper brazing is highly susceptible to corrosion and galvanic attack when exposed to untreated or contaminated water typical in 'legacy commercial radiator systems'.

Suggestion: Specify Nickel-brazed or 100% stainless steel fusion-bonded BPHEs if water quality cannot be strictly controlled.

MANUFACTURABILITY

Applying elastomeric thermal insulation jackets over externally routed trace heating and temperature sensors is labor-intensive and creates air gaps.

Suggestion: Design a custom, pre-molded EPS or EPP multi-piece insulation jacket with integrated channels for sensors and trace heating cables.

ENGINEERING

Deep subcooling using asymmetric geometries to minimize R290 charge is excellent for EN 378 compliance, but highly sensitive to fouling on the water side which shrinks the effective heat transfer area quickly.

Suggestion: Integrate a mandatory Y-strainer or magnetic dirt separator upstream of the BPHE's water inlet.

Strategic Recommendations

- Switch to a Nickel-brazed or fusion-bonded stainless BPHE to withstand legacy system water chemistry.
- Add mechanical anti-freeze valves to physically drain the exchanger during unpowered, sub-zero conditions.
- Replace wrapped elastomeric insulation with a pre-molded, quick-assemble EPP jacket featuring wire routing channels.
- Specify a mandatory inlet water filter/strainer to prevent metallic scaling and fouling from older radiator setups.

4.4 Refrigerant Routing & Expansion Module

Lead Time: 8 weeks

Status: draft

Module Purpose

Regulates R290 flow rate, dictates superheat, and reverses thermodynamic cycle for defrosting.

Why it matters to the system

Protects the compressor by managing precise superheat and enables critical reverse-cycle defrost operations.

Detailed Technical Description

Acting as the hydraulic circulatory center for the refrigerant, this module contains the Electronic Expansion Valve (EEV), 4-way reversing valve, and associated sensing arrays. The EEV provides hundreds of micro-steps to dynamically throttle the liquid R290, flashing it into a low-pressure two-phase state. This rapid active modulation guarantees optimal superheat precisely matched to the inverter compressor's curve, preventing liquid slugging while maximizing efficiency across volatile ambient temperature ranges.

The 4-way valve allows the system to run in reverse, dumping heat from the building back into the outdoor evaporator coil to melt rapid frost buildup. Given R290's flammability, minimizing the total volume of pipework in this module is essential to remain under the EN 378 charge limits. All brazing techniques must meet highly verified non-destructive testing requirements, as this assembly contains the highest quantity of joint failure points in the monobloc unit.

SYSTEM INPUTS	SYSTEM OUTPUTS
- High-pressure subcooled liquid (heating mode) - Microcontroller stepping signals	- Low-pressure flashed liquid/gas mixture - Routed high/low pressure channels

4.4.2 Specifications & Risk Profile

Expected Key Parts

- Electronic Expansion Valve (EEV)
- 4-way reversing valve
- Filter drier
- High/Low pressure transducers
- Copper pipework

Identified Failure Modes

- EEV stepper motor locking in closed position leading to low-pressure trip
- 4-way valve slide sticking mid-position causing high-pressure bypass
- Filter drier core disintegration clogging the EEV orifice

Open Questions

- Reaction time required for EEV adjustments during sudden compressor frequency shifts
- Optimal placement of pressure transducers to avoid turbulent noise in readings

4.4.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
refrigerant_management-001	Electronic Expansion Valve (EEV)	COTS	-	£150.00
refrigerant_management-002	4-way reversing valve	COTS	-	£80.00
refrigerant_management-003	Filter drier	COTS	-	£40.00
refrigerant_management-004	High/Low pressure transducers	COTS	-	£60.00
refrigerant_management-005	Copper pipework	Custom	-	£80.00
Module Total Cost:				£410.00

4.4.4 Engineering Review

Review by Fang

WARN

Verdict Summary

While functionally robust, the high density of manual braze joints combined with an A3 flammable refrigerant (R290) introduces significant manufacturing and safety risks that require rigorous process controls.

Identified Issues

RISK

High concentration of brazed joints poses a severe leak risk for highly flammable R290 refrigerant.

Suggestion: Maximize CNC tube bending to eliminate unnecessary fittings and reduce total braze joints. Mandate 100% Helium Mass Spectrometry leak testing under vacuum for this sub-assembly.

MANUFACTURABILITY

Thermal distortion or internal damage to the 4-way valve and filter drier core during brazing.

Suggestion: Require automated induction brazing with strict temperature limits, or mandate continuous water-cooling wraps for manual brazing of these specific components.

ENGINEERING

Filter drier core disintegration leading to EEV orifice clogging.

Suggestion: Specify a 100% solid-core molecular sieve filter drier with an integrated 20-micron outlet mesh specifically rated for R290 and inverter-driven vibration profiles.

TOLERANCES

Copper pipework routing must be highly compact to stay under EN 378 charge limits.

Suggestion: Implement 3D visual aids and hard-tooled fixturing on the production line to ensure exact pipework geometries are maintained to within +/- 1.0mm.

Strategic Recommendations

- Redesign the tubing layout to consolidate branching and leverage CNC mandrel bending, reducing joint count.
- Implement 100% Helium mass spectrometry leak detection in a localized vacuum chamber for every completed routing module.
- Validate brazing operator qualifications specifically for tight-clearance R290 assemblies, and transition to induction brazing to limit Heat Affected Zones (HAZ).
- Source filter driers with an integrated mechanical screen on the downstream side to prevent any micro-particulates from reaching the EEV.

4.5 Aerodynamic & Airflow Module

Lead Time: 14 weeks

Status: draft

Module Purpose

Drives necessary volumetric airflow through the evaporator coil while silencing operational noise.

Why it matters to the system

Fundamentally dictates the noise profile (dBA) of the unit, which is the primary barrier to securing commercial installation permits.

Detailed Technical Description

To achieve 30kW capability, massive amounts of ambient air must be drawn through the evaporator. This module utilizes specialized large-diameter, low-RPM Electronically Commutated (EC) axial fans to meet these immense volumetric requirements. The composite aero-acoustic blades are specifically molded with trailing edge serrations to manipulate the wake and minimize tonal noise, a critical aspect of meeting the ErP Directive 2009/125/EC for urban settings.

The molded fan cowl dictates the aerodynamic efficiency of the unit; computational fluid dynamics (CFD) are heavily leveraged to prevent tip vortex recirculation and back-pressure. In the event of an internal R290 leak (detected by the safety logic), this module serves a secondary emergency purpose: the fans are immediately driven to maximum RPM to exhaust the heavier-than-air propane gas away from the chassis, diluting it below its Lower Flammability Limit (LFL).

SYSTEM INPUTS	SYSTEM OUTPUTS
- Ambient external air - DC power for EC motors - PWM speed control signals	- Directed high-volume exhaust air - Acoustic emissions

4.5.2 Specifications & Risk Profile

Expected Key Parts

- Electronically Commutated (EC) axial fans
- Molded composite fan blades
- Aerodynamic fan cowl/shroud
- Anti-vibration motor struts
- Wire protection grilles

Identified Failure Modes

- Bearing wear leading to off-axis wobble and catastrophic blade strike on cowl
- PWM signal loss defaulting fan to 0 RPM and triggering high-temp lockout
- Ice build-up on fan blades causing extreme vibrational imbalance

Open Questions

- Interaction of air streams if dual fans are utilized in the monobloc frame
- Actual velocity profile impacts of the safety wire grilles on CFD assumptions

4.5.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
airflow_management-001	Electronically Commutated (EC) axial fans	COTS	-	£250.00
airflow_management-002	Molded composite fan blades	COTS	-	£250.00
airflow_management-003	Aerodynamic fan cowl/shroud	Custom	-	£250.00
airflow_management-004	Anti-vibration motor struts	Custom	-	£80.00
airflow_management-005	Wire protection grilles	Custom	-	£60.00
Module Total Cost:				£890.00

4.5.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The module features advanced aero-acoustic designs that meet high volumetric demands, but presents critical safety risks regarding R290 vapor ignition and requires strict geometric control during the molding process.

Identified Issues

RISK

Using the fan as an emergency R290 exhaust mandates that the EC motor and electrical connections are explosion-proof or intrinsically safe to avoid igniting the flammable mixture.

Suggestion: Specify ATEX-certified or hermetically sealed EC motors, and ensure fan blades use anti-static, non-sparking materials.

ENGINEERING

Defaulting to 0 RPM upon PWM signal loss restricts safety functionality. If a leak and a signal wire failure happen simultaneously, the propane will not be exhausted.

Suggestion: Configure the EC motor firmware to default to 100% RPM upon loss of PWM signal.

TOLERANCES

Minimizing tip vortex recirculation requires very tight gap tolerances between the blade tips and the molded cowl, which conflicts with the 'bearing wear/wobble' failure mode.

Suggestion: Use an abradable liner on the inner diameter of the fan cowl. This allows tight running clearances but safely sacrifices the liner rather than destroying the blade if extreme wobble occurs.

MANUFACTURABILITY

Trailing edge serrations on molded composite blades require highly complex tooling and are vulnerable to flash, sink marks, or warp, which will destroy dynamic balance.

Suggestion: Mandate rigorous mold-flow analysis for tool design, and utilize post-mold cooling fixtures prior to dynamic balancing operations.

MATERIALS

Ice build-up causing severe vibrational imbalance is noted but lacks a mitigation mechanism at the material or hardware level.

Suggestion: Consider a hydrophobic, ice-phobic coating on the composite blades or embedding a low-wattage heating trace in the aerodynamic cowl to prevent freeze-up.

Strategic Recommendations

- Upgrade the entire fan assembly (motor and blades) to meet ATEX or equivalent non-sparking standards for safe R290 emergency purging.
- Invert the PWM fail-safe logic so that a loss of communication defaults the fan to maximum airflow.
- Introduce abradable materials or sacrificial wear-rings on the inside of the fan shroud to prevent catastrophic shattering in the event of a blade strike.
- Implement automated inline dynamic balancing with heavily controlled post-mold cooling sequences for all composite blade assemblies.

4.6 Structural Frame & Base Pan Module

Lead Time: 10 weeks

Status: draft

Module Purpose

Provides rigid support for all internal components, manages condensate runoff, and allows for transport.

Why it matters to the system

Determines the physical footprint for architects mapping out spatial constraints, whilst protecting against long-term environmental degradation.

Detailed Technical Description

Constructed from heavy-gauge powder-coated galvanized steel, the structural chassis acts as the skeleton for the 350kg+ 30kW system. It must be engineered to withstand rigorous commercial transport stresses, utilizing reinforced forklift channels and crane hoisting points to allow installation in tight urban alleyways. Component layout within the frame is spatially optimized to maintain a compact footprint while ensuring strict physical separation between volatile R290 refrigerant zones and sparking electrical contacts.

The base pan performs a crucial environmental function: managing the vast quantities of water generated during the winter defrost cycles. The pan is deep-drawn with complex geometric slopes directing water toward a centralized drain, preventing internal pooling. A dedicated base pan trace heater is incorporated to prevent the condensate from refreezing into a solid block of ice that could eventually encroach on the fan blades or crush bottom-tier piping.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Mechanical loads from internal components - Vibrational energy - Defrost condensate	- Stable geometric foundation - Directed water drainage to ambient

4.6.2 Specifications & Risk Profile

Expected Key Parts

- Powder-coated galvanized steel frame
- Deep-drawn base pan
- Integral forklift/hoist points
- Condensate drainage channel
- Chassis vibration isolators

Identified Failure Modes

- Galvanic corrosion at fastener points breaching paint coating
- Base pan trace heater failure causing solid ice block formation in chassis
- Frame torsion during crane lifting bending delicate copper pipework

Open Questions

- Maximum coastal salt-fog survivability of the chosen powder-coat thickness
- Optimal slope angle in base pan to balance height restrictions with rapid water clearance

4.6.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
structural_chassis-001	Powder-coated galvanized steel frame	Custom	-	£60.00
structural_chassis-002	Deep-drawn base pan	Custom	-	£60.00
structural_chassis-003	Integral forklift/hoist points	Custom	-	£60.00
structural_chassis-004	Condensate drainage channel	Custom	-	£60.00
structural_chassis-005	Chassis vibration isolators	COTS	-	£120.00
Module Total Cost:				£360.00

4.6.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The module appropriately prioritizes ruggedness and R290 safety layout, but deep-drawing a heavy-gauge base pan for a 30kW unit is highly impractical and tooling-intensive. Furthermore, the risk of structural torsion rupturing volatile R290 copper lines during transit requires immediate attention.

Identified Issues

MANUFACTURABILITY

Deep-drawing a heavy-gauge galvanized steel base pan large enough for a 30kW, 350kg system requires colossal press tonnage, risks material tearing, and incurs astronomical tooling costs.

Suggestion: Transition to a stamped and press-braked sheet metal assembly with sealed seams, or utilize a structural steel sub-frame combined with an injection-molded composite/polymer drip tray.

ENGINEERING

Frame torsion during crane lifting is identified as a failure mode that can bend delicate internal copper tubing. A ruptured R290 line due to a lifting operation is a catastrophic safety failure.

Suggestion: Reinforce the chassis utilizing a tubular steel skeleton or diagonal cross-bracing specifically engineered to eliminate torsional deflection under dynamic 4-point lift loads.

MATERIALS

Galvanic corrosion at fastener points is a documented risk, specifically if dissimilar metal fasteners (like stainless steel) cut through the powder coat and galvanization into raw steel.

Suggestion: Mandate dielectric barrier washers (nylon/EPDM) and specify coated fasteners (e.g., Magni or Dacromet) that are galvanically neutral to the base steel.

RISK

Deep-drawn complex slopes in the base pan could unintentionally create low-elevation pockets. Because R290 is heavier than air, leaked gas will pool in these areas, creating an explosion risk.

Suggestion: Ensure the drainage channels and base pan slopes promote downward ventilation of R290 gas to the exterior, strictly avoiding any 'collection bowls'.

ENGINEERING

Trace heater failure leading to ice block expansion can crush piping. Replacement of this heater in the field could be extremely difficult if positioned beneath heavy components.

Suggestion: Design a dedicated slide-out track or external access panel for the trace heater to allow quick field replacement without disassembling the chassis or lifting the unit.

Strategic Recommendations

- Redesign the base pan from a single-piece deep-draw to a folded, welded/sealed structure or a non-structural polymer tray supported by a frame.
- Conduct Finite Element Analysis (FEA) on crane lift scenarios. Add cross-bracing until torsional deflection yields zero strain on internal compressor/copper mounts.
- Draft a fastener specification sheet mandating isolated, galvanically compatible hardware for all chassis penetrations.
- Perform a computational fluid dynamics (CFD) simulation for R290 leak pooling inside the newly sloped base pan to ensure safe natural dispersal.

4.7 Main Electrical & Power Distribution Module

Lead Time: 16 weeks

Status: draft

Module Purpose

Distributes and regulates 3-phase high voltage mains power safely in a highly flammable environment.

Why it matters to the system

The highest safety liability node; electrical arcing paired with an R290 leak is the exact catastrophic scenario safety standards aim to prevent.

Detailed Technical Description

This module receives the 3-phase 400V raw utility power and processes it for distribution via step-down transformers and high-power inverter drives. Because switching high voltages necessarily creates arcs, all contactors, relays, and breakers in this module must be either ATEX-rated non-sparking variants or housed in a hermetically separated, positively pressurized sub-enclosure. This fulfills the strict A3 mitigation criteria required by IEC 60335-2-40.

Significant electromagnetic interference (EMI) and radio frequency interference (RFI) are generated by the heavy inverter switching. The module utilizes heavy-duty line filters and continuous grounding loops to shield adjacent commercial systems (e.g., clinic medical equipment or retail point-of-sale systems). The thermal management of the inverter drive itself is critical; it is often heat-sunk to the water return line or heavily finned to prevent power derating on hot days.

SYSTEM INPUTS

- 3-phase 400V AC building power

SYSTEM OUTPUTS

- Modulated inverted AC for compressor
- DC power for fans and logic
- Regulated voltage lines for pumps/heaters

4.7.2 Specifications & Risk Profile

Expected Key Parts

- Non-sparking / ATEX contactors
- EMI/RFI line filters
- High-power inverter drive module
- Step-down transformers
- Hermetically sealed electrical enclosure

Identified Failure Modes

- Contactor points welding closed failing to shut off compressor
- Inverter IGBT module thermal runaway/failure
- Failure of enclosure seal allowing R290 ingress into spark zones

Open Questions

- Actual peak inrush currents dynamically affecting commercial breaker sizing
- Thermal dissipation requirement of the inverter unit during peak +40°C ambient loads

4.7.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
electrical_power-001	Non-sparking / ATEX contactors	COTS	-	£40.00
electrical_power-002	EMI/RFI line filters	COTS	-	£40.00
electrical_power-003	High-power inverter drive module	COTS	-	£400.00
electrical_power-004	Step-down transformers	COTS	-	£40.00
electrical_power-005	Hermetically sealed electrical enclosure	Custom	-	£120.00
Module Total Cost:				£640.00

4.7.4 Engineering Review

Review by Fang

WARN

Verdict Summary

While the ATEX and IEC 60335-2-40 A3 mitigation strategies are theoretically sound, the proposed positive pressurization and water-line heat sinking introduce severe manufacturing complexities and critical field safety risks.

Identified Issues

RISK

Heat-sinking a 400V inverter to a water return line within a sealed electrical enclosure creates extreme risks of condensation or leaks, which could lead to high-voltage arcing and R290 ignition.

Suggestion: Physically isolate the inverter's water-cooled heat sink to the exterior of the enclosure using a sealed thermal bridge, or rely exclusively on external heavy finning with forced air.

MANUFACTURABILITY

Building a positively pressurized, hermetically sealed macro-enclosure is notoriously difficult in high-volume production, requiring slow pressure-decay leak testing and expensive pneumatic hardware.

Suggestion: Maximize the use of localized potting (encapsulation) and ATEX-certified solid-state relays to eliminate spark sources entirely, rather than attempting to seal the entire enclosure.

MATERIALS

Gaskets and seals used for the enclosure must maintain long-term integrity against thermal cycling and resist degradation from R290 outgassing or liquid contact.

Suggestion: Specify R290-compatible elastomers (such as HNBR or specific Neoprene blends) and avoid standard EPDM, which can swell in the presence of hydrocarbons.

ENGINEERING

Heavy inverter switching requires continuous EMI/RFI grounding loops, which necessitates exposed, unpainted conductive mating surfaces on the enclosure.

Suggestion: Design automated masking into the powder-coating process for all grounding pads and use conductive EMI shielding gaskets at all enclosure seams.

Strategic Recommendations

- Redesign the inverter thermal management system to completely separate any water cooling loops from the high-voltage interior of the enclosure to eliminate condensation and leak risks.
- Perform a cost-benefit analysis on replacing the pressurized enclosure concept with 100% inherently non-sparking/potted ATEX components for better assembly line yield.
- If a sealed enclosure is maintained, integrate automated, multi-axis torque-driving stations and 100% line-side helium or pressure decay leak testing.
- Specify and validate R290-resistant sealing materials for all wire glands, connectors, and door gaskets.

4.8 Control, Safety & Telemetry Module

Lead Time: 20 weeks

Status: draft

Module Purpose

Executes logic for thermal optimization, handles BMS connectivity, and manages urgent R290 leak safety protocols.

Why it matters to the system

Optimizes real-time performance to beat competitor SCOP benchmarks while acting as the final safety backstop against explosive hazards.

Detailed Technical Description

Housing the 'brain' of the machine, this module features a highly customized PCB executing complex thermodynamic envelope mapping, variable-speed algorithms, and building management system (BMS) handshakes. It constantly processes high/low-pressure inputs and EEV superheat states to optimize the envelope in real-time, aggressively chasing the highest part-load efficiencies to hit European SCOP metrics heavily weighted in eco-design directives.

Vitality, this module manages the A3 flammability risk through an independent, hard-wired safety interlock circuit. A highly sensitive catalytic bead R290 leak sensor continuously sniffs the compressor compartment limit. If trace amounts of propane (e.g., 20% of the Lower Flammability Limit) are detected, this module bypasses all software, physically cutting power to the compressor while forcing the EC fans into a maximum-velocity emergency evacuation mode.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Sensor telemetry (temp, pressure, flow) - BMS / Modbus RTU commands - R290 leak detector signals	- Actuator logic (EEV, valves, fans, compressor speed) - Safety interlock triggers - Diagnostic panel outputs

4.8.2 Specifications & Risk Profile

Expected Key Parts

- Main customized PCB controller
- Catalytic bead R290 leak sensor
- Modbus/BACnet interface card
- User / Service HMI screen
- Safety hard-interlock relay

Identified Failure Modes

- R290 leak sensor calibration drift causing false-positive lockdowns
- Corrosion of PCB traces from humid ambient ingress
- Firmware crash resulting in loss of EEV control and liquid slugging

Open Questions

- Longevity of the catalytic bead sensor in harsh urban atmospheric conditions
- Integration depth required by target demographics (Modbus vs BACnet standard)

4.8.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
control_and_safety-001	Main customized PCB controller	COTS	-	£180.00
control_and_safety-002	Catalytic bead R290 leak sensor	COTS	-	£60.00
control_and_safety-003	Modbus/BACnet interface card	COTS	-	£40.00
control_and_safety-004	User / Service HMI screen	COTS	-	£180.00
control_and_safety-005	Safety hard-interlock relay	COTS	-	£40.00
Module Total Cost:				£500.00

4.8.4 Engineering Review

Review by Fang

WARN

Verdict Summary

While the hard-wired safety interlock is conceptually robust, utilizing a catalytic bead sensor introduces severe drift and poisoning risks on the assembly line, and the emergency evacuation protocol requires strict ATEX-rated components.

Identified Issues

RISK

Catalytic bead sensors are highly susceptible to poisoning from silicone vapors and other off-gassing sealants commonly used in HVAC manufacturing and assembly.

Suggestion: Replace the catalytic bead sensor with an NDIR (Non-Dispersive Infrared) R290 sensor, which is immune to chemical poisoning and significantly reduces calibration drift.

ENGINEERING

Activating emergency EC fans and mechanically cutting compressor power during an A3 refrigerant leak introduces potential ignition sources if arcing occurs.

Suggestion: Ensure the safety hard-interlock relay is hermetically sealed or solid-state, and verify that the EC fans are specifically Ex-rated (non-sparking) for Zone 2 explosive atmospheres.

MATERIALS

Standard uncoated PCB assemblies will inevitably degrade due to humid ambient ingress, condensation, and temperature cycling in an outdoor heat pump environment.

Suggestion: Implement an automated conformal coating process (e.g., Parylene or acrylic) for the main PCB and ensure the control module housing meets IP67 standards.

ENGINEERING

Firmware crashes leading to a loss of EEV (Electronic Expansion Valve) control can cause catastrophic liquid slugging in the compressor.

Suggestion: Integrate a hardware watchdog timer and a supercapacitor backup to automatically force the EEV to a closed state independent of the firmware if a crash occurs.

MANUFACTURABILITY

Complex wiring involving high-voltage compressor feeds, low-voltage sensor inputs, and sensitive Modbus/BACnet lines risks signal noise and EMI.

Suggestion: Design the module housing with physically segregated routing channels and utilize shielded cables for the Modbus/BMS interface.

Strategic Recommendations

- Swap catalytic bead sensor to an NDIR sensor to prevent assembly-line poisoning and field drift.
- BOM audit to verify the main interlock relay and emergency EC fans are Ex-rated for A3 refrigerants.
- Add an automated conformal coating step to the PCB manufacturing process.
- Implement a redundant hardware watchdog circuit with capacitor-backed fail-safe EEV closure.
- Segregate high-voltage and low-voltage wiring harnesses in the assembly SOP to prevent EMI.

4.9 Acoustic & Environmental Enclosure Module

Lead Time: 8 weeks

Status: draft

Module Purpose

Provides weather protection, limits acoustic transmission to urban environments, and satisfies EU F-Gas labeling.

Why it matters to the system

Differentiates the product visually from industrial 'ugly' cascading blocks while being the primary defense against rigorous noise complaints.

Detailed Technical Description

Because 30kW units are highly visible when retrofitted in commercial alleyways or exterior walls, the architectural aesthetics of the outer enclosure matter significantly to retail and boutique clinic customers. The enclosure consists of interlocking sheet metal panels lined deeply with high-density, fire-retardant elastomeric acoustic foam. This directly targets high-frequency compressor whine and mid-band fan resonance, essential to passing ErP noise legislation for densely populated areas.

Structurally, the enclosure is strictly regulated by EN 378 and F-Gas 2024/573, requiring specialized louvers that ensure internal air volumes are intrinsically swept to prevent pockets of stagnant R290 pooling. The panels must display massive, standardized flammability warnings, safe exclusion zone radiuses, and dedicated points of access that guide service technicians precisely to the internal components without compromising the atmospheric seal more than necessary.

SYSTEM INPUTS	SYSTEM OUTPUTS
• Radiated noise from internal components • Ambient weather conditions	• Acoustically dampened casing • Regulatory visual compliance • Service access ports

4.9.2 Specifications & Risk Profile

Expected Key Parts

- Acoustic dampening elastomeric foam panels
- Weatherproof aesthetic outer casing handles
- F-Gas / Flammability warning decals
- Quick-release service hatches
- Vented louvers

Identified Failure Modes

- Acoustic foam degrading over time and posing a secondary fire hazard
- Panel rattling at specific compressor frequencies destroying acoustic performance
- UV degradation of critical regulatory safety decals

Open Questions

- Optimal louver geometry to vent potential leaked R290 while blocking direct rain ingress
- Perceived aesthetic value of premium casing variants vs base functional models

4.9.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
outer_enclosure-001	Acoustic dampening elastomeric foam panels	COTS	-	£40.00
outer_enclosure-002	Weatherproof aesthetic outer casing handles	COTS	-	£40.00
outer_enclosure-003	F-Gas / Flammability warning decals	COTS	-	£40.00
outer_enclosure-004	Quick-release service hatches	Custom	-	£60.00
outer_enclosure-005	Vented louvers	Custom	-	£60.00
Module Total Cost:				£240.00

4.9.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The enclosure correctly identifies severe EU regulatory requirements for R290 acoustics and safety, but relies on fundamentally contradictory concepts (e.g., an 'atmospheric seal' coexisting with 'vented louvers') that introduce explosive risks.

Identified Issues

ENGINEERING

Contradiction in airflow design: The description mentions an 'atmospheric seal' while simultaneously specifying 'vented louvers' to sweep R290. Propane (R290) is heavier than air; the enclosure cannot be sealed and must rely on continuous, unpowered bottom-edge ventilation to avert gas pooling.

Suggestion: Eliminate 'atmospheric seal' from the service lid requirements. Design the base and lower 150mm of the unit with passive sweeping louvers and zero lips where heavier-than-air gas can pool.

MATERIALS

Acoustic elastomeric foam can absorb leaked R290 gas if it is open-cell, creating an extreme secondary fire/explosion hazard that negates the 'fire-retardant' rating.

Suggestion: Specify strictly closed-cell elastomeric foam with a metallic or foil facing to prevent gas permeation and moisture absorption.

MATERIALS

UV degradation of critical F-gas and exclusion zone safety decals over a 10-15 year outdoor lifespan will result in regulatory non-compliance.

Suggestion: Replace printed decals with stamped, anodized, or laser-etched aluminum plates riveted to the exterior panels.

ENGINEERING

Interlocking sheet metal spans without sufficient stiffness will inevitably resonate and rattle at specific compressor RPMs, degrading the module's primary acoustic purpose.

Suggestion: Incorporate constrained layer damping (CLD) patches or press-brake stiffening ribs into large unsupported panel sections to shift their natural resonant frequency outside the compressor's operating envelope.

Strategic Recommendations

- Redesign the internal volume bottom pan to have a strictly flat or outward-sloping geometry towards the louvers to prevent R290 pooling.
- Verify the elastomeric foam is closed-cell standard and implement an impermeable foil barrier layer.
- Transition all mandatory F-Gas and EN 378 safety warnings from adhesive decals to riveted metal plates.
- Perform FEA modal analysis on the sheet metal panels against the known compressor frequency map to strategically place stiffening ribs.

4. SYSTEM MODULES & ARCHITECTURE

4.10 Hydraulic Pumping & Delivery Module

Lead Time: 10 weeks Status: draft

Module Purpose

Circulates the building's hydronic water through the heat pump and ensures appropriate thermal mass flow.

Why it matters to the system

Vital for efficient thermal transport out of the unit; without reliable hydronic mass flow, the entire high-pressure R290 cycle quickly locks out.

Detailed Technical Description

This sub-assembly manages the delicate interface between the heat pump's output and the legacy commercial building's water infrastructure. It relies on a high-efficiency, variable-speed (PWM) electronically commutated circulator pump to drive vast amounts of water (routinely up to 5 m³/h) across the condenser BPHE. By dynamically reading the inline vortex flow sensor, the logic varies the pump speed to maintain an exact Delta-T (e.g., 5 Kelvin split), ensuring that the R290 is adequately subcooled to maintain a COP of 3.5.

To safeguard against blockages in the older commercial radiator networks, this module features a heavy-duty mechanical Pressure Relief Valve (PRV) and an automatic air purging valve. A sudden loss of water flow would instantly cause the R290 subsystem to spike to dangerous condensing pressures; thus, flow rate telemetry acts as one of the most immediate, hard-coded safety cut-outs in the machine's architecture.

SYSTEM INPUTS

- Building return water
- PWM signal for pump rate
- 230V AC power

SYSTEM OUTPUTS

- Heated supply water to commercial loop
- Flow rate telemetry
- Water pressure telemetry

4.10.2 Specifications & Risk Profile

Expected Key Parts

- Variable-speed high-efficiency circulator pump
- Vortex flow sensor
- Pressure relief safety valve (PRV)
- DN32/DN40 brass inlet/outlet stubs
- Air purging valve

Identified Failure Modes

- Circulator pump impeller cavitation leading to pump destruction
- Vortex flow sensor fouling from legacy system rust/debris causing false cut-outs
- Air lock in the BPHE circuit causing localized boiling and high-pressure R290 trip

Open Questions

- Maximum back-pressure encountered in typical multi-story small clinic retrofits
- Ideal differential pressure setpoint logic for cascading vs. pure standalone operations

4.10.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
hydraulic_pumping-001	Variable-speed high-efficiency circulator pump	COTS	-	£250.00
hydraulic_pumping-002	Vortex flow sensor	COTS	-	£60.00
hydraulic_pumping-003	Pressure relief safety valve (PRV)	COTS	-	£80.00
hydraulic_pumping-004	DN32/DN40 brass inlet/outlet stubs	Custom	-	£80.00
hydraulic_pumping-005	Air purging valve	COTS	-	£80.00
Module Total Cost:				£550.00

4.10.4 Engineering Review

Review by Fang

FAIL

Verdict Summary

The module lacks essential filtration and magnetic separation required to protect sensitive components (ECM pump, BPHE, vortex sensor) from debris endemic to legacy commercial piping.

Identified Issues

ENGINEERING

ECM pumps utilize powerful permanent magnets that attract magnetite (black iron oxide) from old commercial pipes, leading to rapid bearing and rotor destruction.

Suggestion: Integrate a dual-action magnetic dirt separator at the module's inlet before the pump.

RISK

Legacy pipe debris will clog the BPHE micro-channels and foul the bluff body of the vortex flow sensor, causing false cut-outs and localized boiling.

Suggestion: Add a serviceable Y-strainer (e.g., 60-80 mesh) upstream of the sensor and BPHE.

ENGINEERING

Relying strictly on a single vortex flow sensor telemetry for an immediate R290 high-pressure safety cut-out lacks redundancy, especially given the high fouling risk.

Suggestion: Incorporate a mechanical differential pressure (DP) switch across the BPHE as a fail-safe hardwired cut-out.

MATERIALS

Interfacing DN32/DN40 brass stubs directly to legacy commercial piping (usually black iron or steel) introduces the risk of galvanic corrosion.

Suggestion: Specify dielectric unions or specialized isolating flanges at the building interface points.

ENGINEERING

Pump cavitation is listed as a failure mode, but there is no mechanism to monitor Net Positive Suction Head (NPSH) or minimum system pressure.

Suggestion: Add a low-pressure transducer or cut-out switch on the pump's sub-atmospheric/suction side.

Strategic Recommendations

- Mandate the integration of a magnetic dirt separator and Y-strainer on the return/inlet line to protect the pump and BPHE.
- Add a mechanical DP switch across the BPHE for redundant flow verification.
- Ship the unit with dielectric unions for the DN32/DN40 brass inlet/outlet stubs.
- Include a low water pressure switch to prevent pump operation if the system loses static charge, avoiding cavitation.

5. BILL OF MATERIALS (BOM)

5.1 BOM — Refrigerant Compression Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
compressor_assembly-001	Inverter-driven scroll compressor	purchased_cots / varies	COTS	-	£800.00
compressor_assembly-002	Vibration isolation mounts	purchased_cots / varies	COTS	-	£40.00
compressor_assembly-003	Acoustic compressor jacket	purchased_cots / varies	COTS	-	£800.00
compressor_assembly-004	Discharge temperature sensor	purchased_cots / varies	COTS	-	£60.00
compressor_assembly-005	Sump heater	purchased_cots / varies	COTS	-	£40.00

5. BILL OF MATERIALS (BOM)

5.2 BOM — Air-Side Heat Exchanger Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
air_side_evapo-rator-001	Copper-tube/Aluminum-fin coil bank	purchased_cots / varies	COTS	-	£40.00
air_side_evapo-rator-002	Hydrophilic fin coating	purchased_cots / varies	COTS	-	£40.00
air_side_evapo-rator-003	Refrigerant distribution headers	cnc / varies	Custom	-	£80.00
air_side_evapo-rator-004	Defrost temperature thermistors	purchased_cots / varies	COTS	-	£40.00

5. BILL OF MATERIALS (BOM)

5.3 BOM — Water-Side Heat Exchanger Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
water_side_con- denser-001	Stainless Steel Brazed Plate Heat Exchanger (BPHE)	purchased_cots / varies	COTS	-	£600.00
water_side_con- denser-002	Elastomeric thermal insulation jacket	purchased_cots / varies	COTS	-	£15.00
water_side_con- denser-003	Water-side temperature sensors	purchased_cots / varies	COTS	-	£60.00
water_side_con- denser-004	Freeze protection trace heating	purchased_cots / varies	COTS	-	£40.00

5. BILL OF MATERIALS (BOM)

5.4 BOM — Refrigerant Routing & Expansion Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
refrigerant_man- agement-001	Electronic Expansion Valve (EEV)	purchased_cots / varies	COTS	-	£150.00
refrigerant_man- agement-002	4-way reversing valve	purchased_cots / varies	COTS	-	£80.00
refrigerant_man- agement-003	Filter drier	purchased_cots / varies	COTS	-	£40.00
refrigerant_man- agement-004	High/Low pressure transducers	purchased_cots / varies	COTS	-	£60.00
refrigerant_man- agement-005	Copper pipework	cnc / varies	Custom	-	£80.00

5. BILL OF MATERIALS (BOM)

5.5 BOM — Aerodynamic & Airflow Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
airflow_management-001	Electronically Commutated (EC) axial fans	purchased_cots / varies	COTS	-	£250.00
airflow_management-002	Molded composite fan blades	purchased_cots / varies	COTS	-	£250.00
airflow_management-003	Aerodynamic fan cowl/shroud	sheet_metal / varies	Custom	-	£250.00
airflow_management-004	Anti-vibration motor struts	cnc / varies	Custom	-	£80.00
airflow_management-005	Wire protection grilles	sheet_metal / varies	Custom	-	£60.00

5. BILL OF MATERIALS (BOM)

5.6 BOM — Structural Frame & Base Pan Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
structural_chassis-001	Powder-coated galvanized steelsheet_frame	metal / varies	Custom	-	£60.00
structural_chassis-002	Deep-drawn base pan	sheet_metal / varies	Custom	-	£60.00
structural_chassis-003	Integral forklift/hoist points	sheet_metal / varies	Custom	-	£60.00
structural_chassis-004	Condensate drainage channel	sheet_metal / varies	Custom	-	£60.00
structural_chassis-005	Chassis vibration isolators	purchased_cots / varies	COTS	-	£120.00

5. BILL OF MATERIALS (BOM)

5.7 BOM — Main Electrical & Power Distribution Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
electrical_power-001	Non-sparking / ATEX contactors	purchased_cots / varies	COTS	-	£40.00
electrical_power-002	EMI/RFI line filters	purchased_cots / varies	COTS	-	£40.00
electrical_power-003	High-power inverter drive module	purchased_cots / varies	COTS	-	£400.00
electrical_power-004	Step-down transformers	purchased_cots / varies	COTS	-	£40.00
electrical_power-005	Hermetically sealed electrical enclosure	sheet_metal / varies	Custom	-	£120.00

5. BILL OF MATERIALS (BOM)

5.8 BOM — Control, Safety & Telemetry Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
con-trol_and_safe-ty-001	Main customized PCB controller	purchased_cots / varies	COTS	-	£180.00
con-trol_and_safe-ty-002	Catalytic bead R290 leak sensor	purchased_cots / varies	COTS	-	£60.00
con-trol_and_safe-ty-003	Modbus/BACnet interface card	purchased_cots / varies	COTS	-	£40.00
con-trol_and_safe-ty-004	User / Service HMI screen	purchased_cots / varies	COTS	-	£180.00
con-trol_and_safe-ty-005	Safety hard-interlock relay	purchased_cots / varies	COTS	-	£40.00

5. BILL OF MATERIALS (BOM)

5.9 BOM — Acoustic & Environmental Enclosure Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
outer_enclosure-001	Acoustic dampening elastomer- ic foam panels	purchased_cots / varies	COTS	-	£40.00
outer_enclosure-002	Weatherproof aesthetic outer casing handles	purchased_cots / varies	COTS	-	£40.00
outer_enclosure-003	F-Gas / Flammability warning decals	purchased_cots / varies	COTS	-	£40.00
outer_enclosure-004	Quick-release service hatches	sheet_metal / varies	Custom	-	£60.00
outer_enclosure-005	Vented louvers	sheet_metal / varies	Custom	-	£60.00

5. BILL OF MATERIALS (BOM)

5.10 BOM — Hydraulic Pumping & Delivery Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
hydraulic_pump-ing-001	Variable-speed high-efficiency circulator pump	purchased_cots / varies	COTS	-	£250.00
hydraulic_pump-ing-002	Vortex flow sensor	purchased_cots / varies	COTS	-	£60.00
hydraulic_pump-ing-003	Pressure relief safety valve (PRV)	purchased_cots / varies	COTS	-	£80.00
hydraulic_pump-ing-004	DN32/DN40 brass inlet/outlet stubs	cnc / varies	Custom	-	£80.00
hydraulic_pump-ing-005	Air purging valve	purchased_cots / varies	COTS	-	£80.00

6. Cost Waterfall & Economics

UNIT COST	TARGET CEILING	HEADROOM
£9,367.50	None	N/A

Make vs Buy Split

COTS (BUY) TOTAL	CUSTOM (MAKE) TOTAL	MAKE %
£5,135.00	£1,110.00	11.8%

Per-Module Roll-up

MODULE	% OF UNIT	COST
Module compressor_assembly	27.9%	£2,610.00
Module air_side_evaporator	3.2%	£300.00
Module water_side_condenser	11.4%	£1,072.50
Module refrigerant_management	6.6%	£615.00
Module airflow_management	14.3%	£1,335.00
Module structural_chassis	5.8%	£540.00
Module electrical_power	10.2%	£960.00
Module control_and_safety	8.0%	£750.00
Module outer_enclosure	3.8%	£360.00
Module hydraulic_pumping	8.8%	£825.00

NRE & Overheads

Total NRE (Tooling, Certs)	£6,000.00
Complexity Overhead Multiplier	1.5×

Grand Total (Unit + NRE)

£15,367.50

Cost Confidence Assessment

Costing methodology employs deterministic expansions over a historical parts library combined with spot-price indices for raw materials. Custom parts are calculated volumetrically and adjusted by domain-specific complexity multipliers to ensure realistic unit economics.

Make-versus-buy categorisation is automated based on process availability. Non-recurring engineering (NRE) totals reflect tooling, testing, and compliance costs amortised over the target batch size. Margins of error apply to COTS components subject to global supply chain volatility.

7. RISKS REGISTER (FMEA)

7.1 Refrigerant Compression Module Risks

Risk 1: Internal electrical arcing igniting pooled R290

Base Score: S:5 × L:1 = 5

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Utilize hermetically sealed, intrinsically safe compressor terminal blocks.

Owner: Lead Electrical Engineer

Risk 2: Excessive noise during high-load low-ambient condition

Base Score: S:3 × L:4 = 12

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Develop dynamic phase-shifting in the inverter to smooth torque ripple.

Owner: Systems & Thermal Engineer

7. RISKS REGISTER (FMEA)

7.1 Refrigerant Compression Module Risks

Risk 3: Vibration transmitting to chassis causing pipe fatigue		Base Score: S:4 × L:3 = 12
CAUSE	CONSEQUENCE	
Unknown	Unknown	
EXISTING CONTROLS	PLANNED MITIGATION	
None	Specify dual-stage elastomeric isolation mounts.	
Owner: Mechanical Engineer		

7. RISKS REGISTER (FMEA)

7.2 Air-Side Heat Exchanger Module Risks

Risk 4: Ice expansion deforming or crushing coil fins

Base Score: $S:4 \times L:3 = 12$

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Optimize fin pitch (e.g., 2.5mm+) and enforce maximum defrost interval logic.

Owner: Systems & Thermal Engineer

Risk 5: R290 leak on the air side

Base Score: $S:4 \times L:2 = 8$

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Execute automated helium leak testing during module sub-assembly.

Owner: Manufacturing Engineer

7. RISKS REGISTER (FMEA)

7.2 Air-Side Heat Exchanger Module Risks

Risk 6: Uneven refrigerant distribution leading to capacity loss		Base Score: S:3 × L:3 = 9
CAUSE	CONSEQUENCE	
Unknown	Unknown	
EXISTING CONTROLS	PLANNED MITIGATION	
None	Use CFD to optimize header distributor tube lengths and cone depths.	
Owner: Thermal Engineer		

7. RISKS REGISTER (FMEA)

7.3 Water-Side Heat Exchanger Module Risks

Risk 7: BPHE freeze-rupture mixing R290 into building water loop

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Include primary and secondary freeze-stat logic plus fail-safe trace heating.

Owner: Systems & Thermal Engineer

Risk 8: Scale buildup limiting maximum 75°C output

Base Score: S:3 × L:4 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Require inline magnetic/dirt filters in installation manuals and design high-turbulence channels.

Owner: Mechanical/Piping Engineer

7. RISKS REGISTER (FMEA)

7.3 Water-Side Heat Exchanger Module Risks

Risk 9: Loss of certification for PED 2014/68/EU

Base Score: S:5 × L:1 = 5

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Strict vendor auditing and mandatory traceability certs on all BPHE units.

Owner: Compliance & Documentation Manager

7. RISKS REGISTER (FMEA)

7.4 Refrigerant Routing & Expansion Module Risks

Risk 10: Brazed joint fracture leading to catastrophic R290 dump

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Implement automated orbital brazing protocols and 100% helium sniffing on pipework sub-assemblies.

Owner: Mechanical/Piping Engineer

Risk 11: 4-way valve mechanical failure during winter

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Specify pilot-operated bi-stable valves capable of operating under high delta-pressure.

Owner: Lead Mechanical Engineer

7. RISKS REGISTER (FMEA)

7.4 Refrigerant Routing & Expansion Module Risks

Risk 12: Expansion noise radiating outside chassis

Base Score: S:2 × L:4 = 8

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Wrap EEV and immediate pipework in heavy elastomeric acoustic lagging.

Owner: Acoustic Engineer

7. RISKS REGISTER (FMEA)

7.5 Aerodynamic & Airflow Module Risks

Risk 13: Exceeding ErP Lot 1 acoustic limits at high fan load

Base Score: S:4 × L:3 = 12

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Oversize fan diameter to allow lower RPM for equivalent CFM displacement.

Owner: Systems & Thermal Engineer

Risk 14: Fan ignition of leaked R290

Base Score: S:5 × L:1 = 5

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Source strictly ATEX Zone 2 rated EC fan motors with sealed electronics.

Owner: Lead Electrical Engineer

7. RISKS REGISTER (FMEA)

7.5 Aerodynamic & Airflow Module Risks

Risk 15: Vibration transmitting through motor struts		Base Score: S:3 × L:4 = 12
CAUSE	CONSEQUENCE	
Unknown	Unknown	
EXISTING CONTROLS	PLANNED MITIGATION	
None	Isolate motor struts from main chassis using custom rubber grommets.	
Owner: Mechanical Engineer		

7. RISKS REGISTER (FMEA)

7.6 Structural Frame & Base Pan Module Risks

Risk 16: Corrosive failure of base pan leading to structural collapse

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Utilize marine-grade aluminum or apply secondary epoxy coating for the base pan.

Owner: Mechanical Engineer

Risk 17: Crane hoist points failing under heavy shock load

Base Score: S:5 × L:1 = 5

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

FEA simulation of 3G dynamic loading on lifting points; physical drop validation.

Owner: Manufacturing Engineer

7. RISKS REGISTER (FMEA)

7.6 Structural Frame & Base Pan Module Risks

Risk 18: Defrost water freezing on alleyway pavements creating slip hazard		Base Score: S:3 × L:5 = 15
CAUSE	CONSEQUENCE	
Unknown	Unknown	
EXISTING CONTROLS	PLANNED MITIGATION	
None	Design standard DN40 stub-out for directing runoff into local building drainage systems.	
Owner: Mechanical/Piping Engineer		

7. RISKS REGISTER (FMEA)

7.7 Main Electrical & Power Distribution Module Risks

Risk 19: R290 ignition via relay spark

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Complete physical/hermetic segregation of the electrical housing from the compressor bay.

Owner: Lead Electrical Engineer

Risk 20: Inverter overheat shutting down entire heat pump

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Water-cooled cold plate for inverter IGBTs utilizing re-turn-line hydronics.

Owner: Systems & Thermal Engineer

7. RISKS REGISTER (FMEA)

7.7 Main Electrical & Power Distribution Module Risks

Risk 21: Harmonics pollution to building grid

Base Score: S:3 × L:3 = 9

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Implement heavy active harmonic filtering and comprehensive CE-EMC testing.

Owner: Lead Electrical Engineer

7. RISKS REGISTER (FMEA)

7.8 Control, Safety & Telemetry Module Risks

Risk 22: Sensor drift disables R290 safety evacuation mechanism

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Utilize dual-redundant infrared (NDIR) sensors with auto-mated self-calibration diagnostics.

Owner: Lead Electrical Engineer

Risk 23: Microcontroller lockup freezing defrost timer

Base Score: S:3 × L:3 = 9

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Implement a hardware watchdog timer to force-reboot the CPU upon freeze.

Owner: Software/Controls Engineer

7. RISKS REGISTER (FMEA)

7.8 Control, Safety & Telemetry Module Risks

Risk 24: Moisture shorting out low voltage logic rails

Base Score: S:4 × L:3 = 12

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Apply heavy conformal coating to all PCBs and house in IP65 enclosure.

Owner: Manufacturing Engineer

7. RISKS REGISTER (FMEA)

7.9 Acoustic & Environmental Enclosure Module Risks

Risk 25: Vented louvers allowing snow build-up inside chassis

Base Score: S:3 × L:4 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Design labyrinthine vent paths and internal baffles.

Owner: Mechanical Engineer

Risk 26: Acoustic foam soaking up water/oil and degrading

Base Score: S:3 × L:3 = 9

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Specify closed-cell acoustic foam with oleophobic facing.

Owner: Systems & Thermal Engineer

7.9 Acoustic & Environmental Enclosure Module Risks

Base Score: S:5 × L:1 = 5

CONSEQUENCE

Unknown

PLANNED MITIGATION

Incorporate decal application into final QA vision system check.

This section was produced by google/gemini-3.1-pro-preview via OpenRouter. Source: LLM research synthesis.

7. RISKS REGISTER (FMEA)

7.10 Hydraulic Pumping & Delivery Module Risks

Risk 28: Total loss of flow triggering R290 high-pressure rapid spike

Base Score: S:4 × L:3 = 12

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Implement ultra-fast polling on the flow sensor tied directly to compressor hard-stop.
Owner: Software/Controls Engineer	

Risk 29: Pump seal failure leading to internal chassis flooding

Base Score: S:3 × L:2 = 6

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Specify commercial-grade ceramic seals and isolate pump module from electronics.
Owner: Mechanical/Piping Engineer	

7. RISKS REGISTER (FMEA)

7.10 Hydraulic Pumping & Delivery Module Risks

Risk 30: Blockage of flow sensor due to poor building water quality		Base Score: S:3 × L:4 = 12
CAUSE	CONSEQUENCE	
Unknown	Unknown	
EXISTING CONTROLS	PLANNED MITIGATION	
None	Integrate a robust Y-strainer/magnetic filter immediately prior to the flow sensor array.	
Owner: Lead Mechanical Engineer		

8. SUPPLIER SHORTLIST & SOURCING

8.1 Inverter-driven scroll compressor (compressor_assembly-001)

8. SUPPLIER SHORTLIST & SOURCING

8.2 Vibration isolation mounts (compressor_assembly-002)

8. SUPPLIER SHORTLIST & SOURCING

8.3 Acoustic compressor jacket (compressor_assembly-003)

8. SUPPLIER SHORTLIST & SOURCING

8.4 Discharge temperature sensor (compressor_assembly-004)

8. SUPPLIER SHORTLIST & SOURCING

8.5 Sump heater (compressor_assembly-005)

8. SUPPLIER SHORTLIST & SOURCING

8.6 Copper-tube/Aluminum-fin coil bank (air_side_evaporator-001)

8. SUPPLIER SHORTLIST & SOURCING

8.7 Hydrophilic fin coating (air_side_evaporator-002)

8. SUPPLIER SHORTLIST & SOURCING

8.8 Refrigerant distribution headers (air_side_evaporator-003)

8. SUPPLIER SHORTLIST & SOURCING

8.9 Defrost temperature thermistors (air_side_evaporator-004)

8. SUPPLIER SHORTLIST & SOURCING

8.10 Stainless Steel Brazed Plate Heat Exchanger (BPHE) (water_side_condenser-001)

8. SUPPLIER SHORTLIST & SOURCING

8.11 Elastomeric thermal insulation jacket (water_side_condenser-002)

8. SUPPLIER SHORTLIST & SOURCING

8.12 Water-side temperature sensors (water_side_condenser-003)

8. SUPPLIER SHORTLIST & SOURCING

8.13 Freeze protection trace heating (water_side_condenser-004)

8. SUPPLIER SHORTLIST & SOURCING

8.14 Electronic Expansion Valve (EEV) (refrigerant_management-001)

8. SUPPLIER SHORTLIST & SOURCING

8.15 4-way reversing valve (refrigerant_management-002)

8. SUPPLIER SHORTLIST & SOURCING

8.16 Filter drier (refrigerant_management-003)

8. SUPPLIER SHORTLIST & SOURCING

8.17 High/Low pressure transducers (refrigerant_management-004)

8. SUPPLIER SHORTLIST & SOURCING

8.18 Copper pipework (refrigerant_management-005)

8. SUPPLIER SHORTLIST & SOURCING

8.19 Electronically Commutated (EC) axial fans (airflow_management-001)

8. SUPPLIER SHORTLIST & SOURCING

8.20 Molded composite fan blades (airflow_management-002)

8. SUPPLIER SHORTLIST & SOURCING

8.21 Aerodynamic fan cowl/shroud (airflow_management-003)

8. SUPPLIER SHORTLIST & SOURCING

8.22 Anti-vibration motor struts (airflow_management-004)

8. SUPPLIER SHORTLIST & SOURCING

8.23 Wire protection grilles (airflow_management-005)

8. SUPPLIER SHORTLIST & SOURCING

8.24 Powder-coated galvanized steel frame (structural_chassis-001)

8. SUPPLIER SHORTLIST & SOURCING

8.25 Deep-drawn base pan (structural_chassis-002)

8. SUPPLIER SHORTLIST & SOURCING

8.26 Integral forklift/hoist points (structural_chassis-003)

8. SUPPLIER SHORTLIST & SOURCING

8.27 Condensate drainage channel (structural_chassis-004)

8. SUPPLIER SHORTLIST & SOURCING

8.28 Chassis vibration isolators (structural_chassis-005)

8. SUPPLIER SHORTLIST & SOURCING

8.29 Non-sparking / ATEX contactors (electrical_power-001)

8. SUPPLIER SHORTLIST & SOURCING

8.30 EMI/RFI line filters (electrical_power-002)

8. SUPPLIER SHORTLIST & SOURCING

8.31 High-power inverter drive module (electrical_power-003)

8. SUPPLIER SHORTLIST & SOURCING

8.32 Step-down transformers (electrical_power-004)

8. SUPPLIER SHORTLIST & SOURCING

8.33 Hermetically sealed electrical enclosure (electrical_power-005)

8. SUPPLIER SHORTLIST & SOURCING

8.34 Main customized PCB controller (control_and_safety-001)

8. SUPPLIER SHORTLIST & SOURCING

8.35 Catalytic bead R290 leak sensor (control_and_safety-002)

8. SUPPLIER SHORTLIST & SOURCING

8.36 Modbus/BACnet interface card (control_and_safety-003)

8. SUPPLIER SHORTLIST & SOURCING

8.37 User / Service HMI screen (control_and_safety-004)

8. SUPPLIER SHORTLIST & SOURCING

8.38 Safety hard-interlock relay (control_and_safety-005)

8. SUPPLIER SHORTLIST & SOURCING

8.39 Acoustic dampening elastomeric foam panels (outer_enclosure-001)

8. SUPPLIER SHORTLIST & SOURCING

8.40 Weatherproof aesthetic outer casing handles (outer_enclosure-002)

8. SUPPLIER SHORTLIST & SOURCING

8.41 F-Gas / Flammability warning decals (outer_enclosure-003)

8. SUPPLIER SHORTLIST & SOURCING

8.42 Quick-release service hatches (outer_enclosure-004)

8. SUPPLIER SHORTLIST & SOURCING

8.43 Vented louvers (outer_enclosure-005)

8. SUPPLIER SHORTLIST & SOURCING

8.44 Variable-speed high-efficiency circulator pump (hydraulic_pump-ing-001)

8. SUPPLIER SHORTLIST & SOURCING

8.45 Vortex flow sensor (hydraulic_pumping-002)

8. SUPPLIER SHORTLIST & SOURCING

8.46 Pressure relief safety valve (PRV) (hydraulic_pumping-003)

8. SUPPLIER SHORTLIST & SOURCING

8.47 DN32/DN40 brass inlet/outlet stubs (hydraulic_pumping-004)

8. SUPPLIER SHORTLIST & SOURCING

8.48 Air purging valve (hydraulic_pumping-005)

9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	local-1777891131920
Research Result	Generated
System Modules	10 Modules
BOM Lines	48 Parts
Cost Analysis	Complete
Specialist Reviews	10 Completed
Supplier Shortlists	48 Matched

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction
Modules	LLM	Unknown LLM	Gemini 3.1 Pro — module decomposition
BOM	DETERMINISTIC	deterministic	Deterministic expansion from Max keyParts
BOM	LLM	Unknown LLM	Gemini 3.1 Pro — gap-fill for missing standard hardware
Cost	DETERMINISTIC	deterministic	Domain overhead multiplier model
Suppliers	SEARCH	search	Brave Search API — commercial supplier matching
Reviews	LLM	Unknown LLM	Gemini 3.1 Pro — Fang engineering review
Proofreader	LLM	Unknown LLM	Gemini 3.1 Pro — cross-module consistency check

11. Quality Scores

Automated self-critique mechanism evaluating the completeness, realism, and engineering validity of each section.

Brief

Score: 8/10

EVALUATION REASONS

- Use case is detailed
- Mission statement present
- Target customers defined
- Market timing rationale present

IMPROVEMENT SUGGESTIONS

- Declare cost ceiling
- Declare mass constraint
- Add compliance notes

Regulatory

Score: 6/10

EVALUATION REASONS

- 5 standards identified
- Most standards have design impact analysis
- Gap actions defined for some standards

IMPROVEMENT SUGGESTIONS

- Add applicability detail to more standards

Sizing

Score: 6/10

EVALUATION REASONS

- Design is infeasible — shown as exception
- Module dimensions computed
- 1 conflicts identified
- Recommendations provided

Modules

Score: 9/10

EVALUATION REASONS

- 10 modules — good granularity
- Most modules have detailed descriptions
- Most modules have 3+ key parts
- Most modules have risk matrices
- 10/10 modules have engineering reviews

BOM

Score: 8/10

EVALUATION REASONS

- 48 parts — comprehensive BOM
- Most parts have cost estimates
- Parts are linked to modules
- Total BOM cost: £6245

IMPROVEMENT SUGGESTIONS

- More parts need mass estimates

Cost

Score: 8/10

EVALUATION REASONS

- Unit cost: £9368
- Per-module breakdown available
- NRE estimated
- Arithmetic closes within 1%

Risks

Score: 8/10

EVALUATION REASONS

- 30 risks identified — comprehensive
- Most risks have mitigations
- Most risks have owners assigned
- 17 high-priority risks (score ≥ 10)

Suppliers

Score: 4/10

EVALUATION REASONS

- Most suppliers have match scores

IMPROVEMENT SUGGESTIONS

- Need more supplier matches
- More parts need supplier matches

Research

Score: 7/10

EVALUATION REASONS

- Research report: 1908 chars
- 5 sources cited
- Domain detected: hvac_commercial_heating